

Linguistic Agency and Emotional Framing in Patient and Caregiver Narratives: A Corpus-Assisted Discourse Analysis

1. Background

1.1 Narrative Medicine and Illness Narratives

Medical narratives provide insight into how patients and caregivers make sense of illness and construct personal experiences. Narrative medicine emphasizes the importance of stories in understanding suffering, identity, and coping (Charon, 2006).

1.2 Linguistic Agency

Agency refers to the degree to which speakers present themselves as active participants or as experiencers of events. Systemic Functional Linguistics (Halliday, 1994) and discourse studies have shown that grammatical choices, such as active and passive voice, subject roles, and modal constructions, shape perceptions of control and responsibility.

1.3 Emotional Framing

Emotions in discourse are expressed through sentiment, evaluative language, and affective vocabulary. Previous studies have examined emotional language in illness narratives, but little attention has been paid to how emotional framing interacts with linguistic agency.

1.4 Research Gap

Although agency and emotion have been investigated separately, few studies have explored their relationship in medical narratives. Moreover, differences between patient and caregiver narratives remain underexplored.

Research Questions

RQ1. How do patient and caregiver narratives differ in their use of linguistic agency?

RQ2. How do emotional patterns differ between patient and caregiver narratives?

RQ3. To what extent are agency-related constructions associated with emotional framing?

Hypotheses

H1. Patient narratives will contain a higher proportion of passive constructions and lower linguistic agency than caregiver narratives.

H2. Patient narratives will display more negative sentiment and stronger emotional intensity.

H3. Reduced agency will be associated with stronger negative emotional framing.

2. Method

Corpus

The corpus consists of 101 English-language medical narratives published in medical journals.

- 51 patient narratives
- 50 caregiver narratives

The corpus contains approximately 6900 sentences.

Materials

Narratives were collected from English-language medical journals containing first-person accounts. Only autobiographical narratives written by patients or caregivers were included.

Linguistic Features

Agency-related features:

- active and passive voice
- agency score
- first-person references
- agency verbs

Emotional features:

- sentiment polarity
- emotional intensity
- positive sentiment
- negative sentiment
- density of affective vocabulary

Procedure

1. Text collection and preprocessing.
2. Sentence segmentation.
3. Extraction of agency-related features.
4. Sentiment analysis.
5. Statistical analysis.

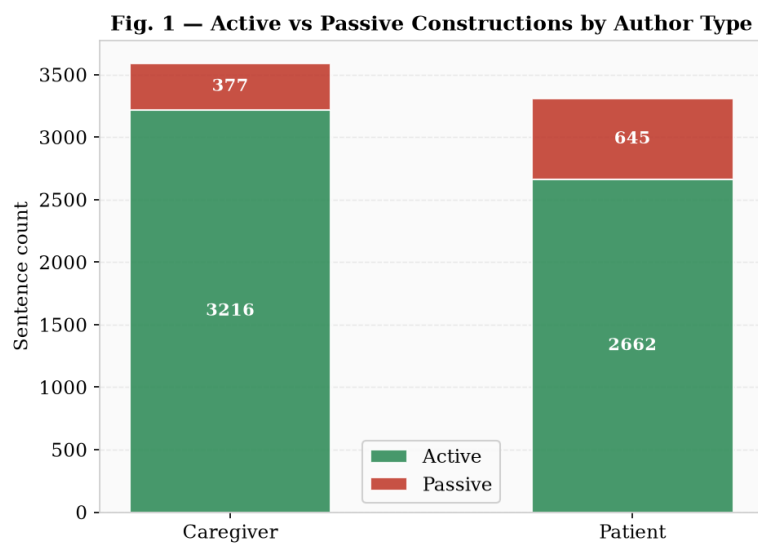
Statistical Analysis

- Mann–Whitney U tests
 - Spearman correlations
 - Descriptive statistics
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3. Results

3.1 Distribution of Agency Constructions

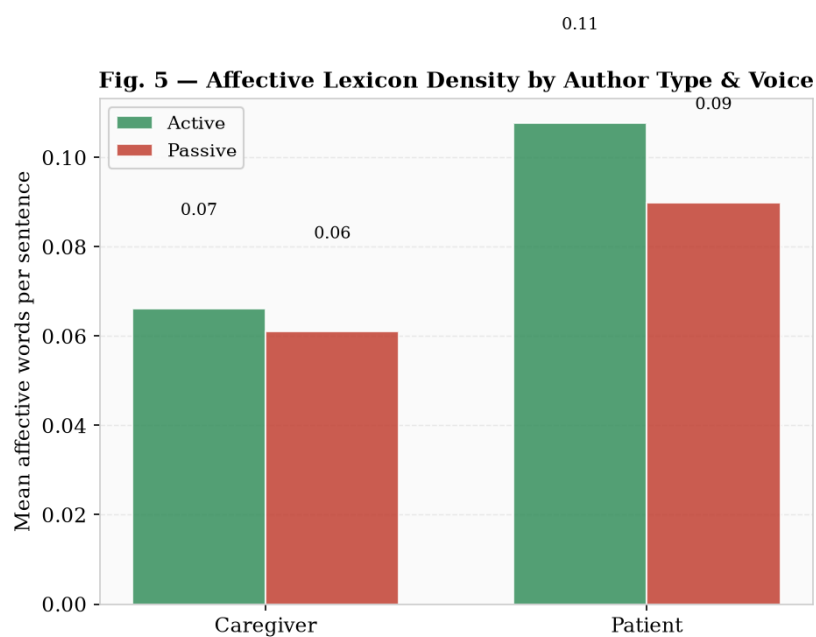
Caregiver narratives contained predominantly active constructions (89%), whereas patient narratives exhibited substantially higher frequencies of passive constructions (21%).



3.2 Emotional Differences

Patient narratives showed:

- lower compound sentiment scores;
- greater emotional intensity;
- more negative affect;
- higher affective density.



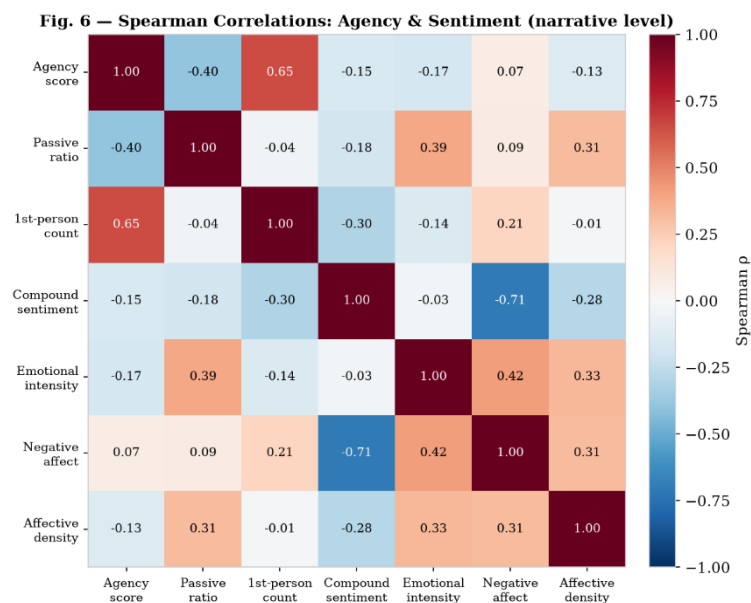
3.3 Agency and Emotion

Passive constructions were associated with slightly stronger emotional intensity and more negative sentiment. However, effect sizes were small.



3.4 Correlation Analysis

Correlations between overall agency scores and emotional measures were weak and mostly non-significant, suggesting that linguistic agency alone does not strongly predict emotional framing.



4. Discussion

The findings partially support the hypothesis that reduced agency is associated with more negative emotional framing. However, this relationship appears to be weak.

More substantial differences emerged between patients and caregivers. Patient narratives were characterized by:

- more passive constructions;
- stronger emotional intensity;
- higher affective density;
- more negative sentiment.

These findings suggest that narrative perspective may be more influential than agency itself.

Patients frequently describe experiences in which events happen to them:

- I was diagnosed.
- I was admitted.
- I was told.

In contrast, caregivers more often present themselves as active agents:

- I helped.
- I called the doctor.
- I cared for her.

Thus, differences in emotional framing may reflect differing social roles and experiences rather than grammatical agency alone.

Limitations

- relatively small corpus;
- automatic sentiment analysis;
- lack of contextual annotation;
- limited range of agency measures.

Future Directions

Future studies could incorporate:

- Appraisal Theory;
- manual discourse annotation;
- topic modelling;
- longitudinal analysis;
- larger corpora.

5. Resources

Primary Sources

Patient and caregiver narratives published in:

- BMJ
- JAMA
- The Lancet
- CMAJ

Software and Tools

- Python
- pandas
- spaCy
- NLTK
- VADER sentiment analyzer
- scipy

Theoretical Resources

Halliday, M. A. K. (1994). *An Introduction to Functional Grammar*.

Charon, R. (2006). *Narrative Medicine: Honoring the Stories of Illness*.

Martin, J. R., & White, P. R. R. (2005). *The Language of Evaluation: Appraisal in English*.

Hyland, K. (2005). *Metadiscourse: Exploring Interaction in Writing*.

Biber, D. (1998). *Corpus Linguistics: Investigating Language Structure and Use*.